Final Project Report: Multi-Dimensional Sales Analysis

1. Project Overview

The objective of this project was to analyze a complex sales dataset to identify patterns in business growth and regional performance. Despite various missing values and data inconsistencies, the project utilized advanced data cleaning and visualization techniques to generate actionable business insights.

2. Dataset Key Features

The dataset consists of 14 columns covering multiple dimensions of sales operations:Identities: Product_ID, Sales_Rep, Region.

Transaction Details: Sale_Date, Sales_Channel, Payment_Method.

Financial Metrics: Sales_Amount, Quantity_Sold, Unit_Cost, Unit_Price, Discount.

3. Data Cleaning Strategy

To maintain the integrity of the analysis, the following technical preprocessing steps were implemented:

A. Missing Value Imputation (Numerical)

Sales_Amount & Quantity_Sold: Due to the presence of outliers (extreme high/low sales) that could skew the average, missing values were filled using the Median.

Unit_Cost: Where cost data was unavailable, a category-wise median was used to maintain realistic margins.

B. Categorical Handling (Text)

Sales_Rep & Region: These were filled with the placeholder "Unknown" to ensure no transaction data was lost while maintaining transparency in reporting.

Product_Category: Missing entries were labeled as "Uncategorized."

C. Logic-Based Correction

Region_and_Sales_Rep: This composite column was used to reconstruct and verify missing entries in the individual Region and Sales_Rep columns.

4. Analytical Insights & Visualizations

A. Regional Performance Analysis

A vertical bar chart was utilized to rank regions by total revenue.

Insight: Identifying top-performing and under-performing territories allows for the strategic allocation of marketing resources and sales staff.

B. Product Category & Inventory Mapping

A heatmap was generated to visualize the correlation between Product Categories and Regions.

Insight: The analysis highlighted that certain categories dominate specific regions, suggesting a need for localized inventory optimization.

C. Sales Channel Efficiency

Comparative bar charts were used to analyze Online vs. In-Store performance.

Insight: The data suggests specific channels respond more effectively to discounts, providing a roadmap for future promotional campaigns.

5. Technical Implementation

The analysis was executed using a standard Python Data Science stack:Data Processing: Pandas and NumPy for manipulation and cleaning.Visualizations: Seaborn (for Heatmaps and Boxplots) and Matplotlib (for customized Bar and Line charts).

Environment: Jupyter Notebook.

6. Business Recommendations

Market Focus: Increase localized discounts and sales representative training in under-performing regions to bridge the revenue gap.

Inventory Optimization: Adjust stock levels based on the high-selling categories identified in the regional heatmap.

Data Quality Control: Implement digital validation at the point of data entry to minimize missing values in future datasets.